

Data Mining Chapter 2 Statistical Learning

Question Group 1

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Question 1

For each of parts (a) through (d), indicate whether we would generally expect the performance of a flexible statistical learning method to be better or worse than an inflexible method. Justify your answer.

a) The sample size n is extremely large, and the number of predictors p is small.

Answer

The flexible model would perform better because the sample size is very large to fit more parameters while preventing overfitting as the small number of predictors limits variance.

b) The number of predictors p is extremely large, and the number of observations n is small.

Answer

The inflexible model would perform better because the flexible model would likely overfit the data.

c) The relationship between the predictors and response is highly non-linear.

Answer

The flexible model would perform better because it will capture the variance present in the data and it is less restrictive on the shape of fit.

d) The variance of the error terms, i.e. $\sigma^2 = Var(\epsilon)$, is extremely high.

Answer

The inflexible model would perform better because the flexible model would capture the variance in the data that occur due to random chance and cause overfitting.

Question 2

This exercise involves the Boston housing data set.

a) To begin, load in the Boston data set, which is part of the ISLP library

Answer

Loading the Boston Data Set

The Boston data set was loaded with:

```
from ISLP import load_data
Boston = load_data('boston')
```

The data were then inspected using `Boston.columns()` and `Boston.info()`.

b) How many rows are in this data set? How many columns? What do the rows and columns represent?

Answer

Number of Rows, Columns, and Their Meaning

The data set contains **506 rows and 14 columns**. Each row represents a Boston suburb/census tract, while the columns describe characteristics of that suburb and its housing market.

Variable	Description
crim	Per-capita crime rate by town.
zn	Proportion of residential land zoned for lots larger than 25,000 square feet.
indus	Proportion of non-retail business acres per town.
chas	Charles River dummy variable: 1 if the tract bounds the river and 0 otherwise.
nox	Nitrogen oxides concentration.
rm	Average number of rooms per dwelling.
age	Proportion of owner-occupied units built before 1940.
dis	Weighted mean distance to five Boston employment centres.
rad	Index of accessibility to radial highways.
tax	Full-value property-tax rate per \$10,000.
ptratio	Pupil-teacher ratio by town.
black	Variable based on the proportion of Black residents, defined in the original data set as $1000(B_k - 0.63)^2$.
lstat	Percentage of the population classified as lower status.
medv	Median value of owner-occupied homes, in thousands of dollars.

c) Make some pairwise scatterplots of the predictors (columns) in this data set. Describe your findings.

Answer

Pairwise Scatterplots of the Predictors

A pairwise scatterplot matrix was produced using Seaborn:

```
sns.pairplot(data=Boston)
```

A correlation heatmap was also used to support interpretation:

```
plt.figure(figsize=[12,10])  
sns.heatmap(Boston.corr(), annot=True);
```

The main findings noted from the notebook are that the per-capita crime rate (**crim**) is related to several of the other variables. In particular, **crim** tends to have a negative relationship with **dis** and **black**, while it has positive relationships with **indus** and **nox**. The relationship between crime and median house value (**medv**) is also negative. These patterns suggest that areas with greater industrial activity and higher nitrogen-oxide concentrations tend to be associated with higher crime rates, whereas areas with characteristics associated with greater distance from employment centres tend to show lower crime rates.

d) Are any of the predictors associated with per capita crime rate? If so, explain the relationship.

Answer

Predictors Associated with Per-Capita Crime Rate

Yes. Several predictors appear to be associated with **crim**. The correlation analysis in the notebook indicates:

- **crim** has a positive relationship with **indus**;
- **crim** has a positive relationship with **nox**;
- **crim** has a negative relationship with **dis**;
- **crim** has a negative relationship with **black**;
- **crim** also tends to be negatively associated with **medv**.

The relationships should be interpreted as associations rather than causal effects. For example, the positive relationship between crime and industrial land use does not by itself establish that industrial activity causes crime.

e) Do any of the suburbs of Boston appear to have particularly high crime rates? Tax rates? Pupil-teacher ratios? Comment on the range of each predictor.

Answer

Particularly High Crime Rates, Tax Rates, and Pupil-Teacher Ratios

The notebook examined the distributions of `crim`, `tax`, and `ptratio` using histograms and selected unusually high observations.

Crime rate

The notebook selected observations with:

$$\text{crim} > 20.$$

There are **18 suburbs** above this threshold. The crime distribution is strongly right-skewed: most suburbs have relatively low crime rates, while a small number have exceptionally high values. The overall crime-rate range is approximately

$$0.00632 \leq \text{crim} \leq 88.9762.$$

Thus, the observations above 20 are clearly in the upper tail.

Tax rate

The notebook selected:

$$\text{tax} > 700.$$

There are **5 suburbs** above this threshold. The overall tax-rate range is approximately

$$187 \leq \text{tax} \leq 711.$$

Therefore, values above 700 represent the extreme upper end of the observed tax-rate distribution.

Pupil-teacher ratio

The notebook examined the distribution of `ptratio` and considered values above 20. The overall range is approximately

$$12.6 \leq \text{ptratio} \leq 22.0.$$

The histogram shows a concentration around the higher end of the distribution, with a noticeable spike near a ratio of 20. The upper-tail values therefore represent suburbs with relatively high pupil-teacher ratios.

f) How many of the suburbs in this data set bound the Charles river?

Answer

(f) Suburbs Bounding the Charles River

The variable `chas` is equal to 1 when a tract bounds the Charles River. The notebook found:

35 suburbs

that bound the Charles River.

g) What is the median pupil-teacher ratio among the towns in this data set?

Answer

Median Pupil-Teacher Ratio

The notebook calculated the median using:

```
np.median(Boston['ptratio'])
```

The median pupil-teacher ratio is:

19.05.

Thus, half of the suburbs have pupil-teacher ratios below 19.05 and half have ratios above 19.05.

h) Which suburb of Boston has lowest median value of owner occupied homes? What are the values of the other predictors for that suburb, and how do those values compare to the overall ranges for those predictors? Comment on your findings.

Answer

Suburbs with the Lowest Median Home Value

The notebook identified observations satisfying:

```
Boston[Boston['medv'] == np.min(Boston['medv'])]
```

The minimum median home value is **\$5,000**, since `medv` is measured in thousands of dollars. There are **two suburbs/census tracts** with this minimum value: observations **399 and 406**.

	crim	zn	indus	chas	nox	rm	age	dis
399	38.3518	0	18.1	0	0.693	5.453	100	1.4896
406	67.9208	0	18.1	0	0.693	5.683	100	1.4254

	rad	tax	ptratio	black	lstat	medv
399	24	666	20.2	396.90	30.59	5
406	24	666	20.2	384.97	22.98	5

Several characteristics of these two suburbs stand out when compared with the overall data range. Both have very high crime rates, very high industrial land use, high nitrogen-oxide concentration, the maximum value of `age`, very short distances to the employment centres, and the maximum value of `rad`. Their tax rate and pupil-teacher ratio are also near the upper end of their respective distributions. In contrast, their average number of rooms per dwelling is low. Both also have high `lstat` values, which is consistent with the very low median home values.

Overall, the two lowest-value suburbs combine several characteristics that are unfavourable from a housing-market perspective: high crime, high industrial activity, high pollution, older housing stock, relatively high taxes, and a high proportion of lower-status population.

i) In this data set, how many of the suburbs average more than seven rooms per dwelling? More than eight rooms per dwelling? Comment on the suburbs that average more than eight rooms per dwelling.

Answer

Suburbs Averaging More Than Seven and Eight Rooms

The number of suburbs with more than seven rooms per dwelling is:

64.

The number with more than eight rooms per dwelling is:

13.

Thus, only a relatively small subset of the 506 suburbs averages more than eight rooms per dwelling.

The 13 suburbs with more than eight rooms tend to have several distinctive characteristics. Their crime rates are generally low relative to the full sample, and many have relatively low property-tax rates. They also tend to have higher housing values and lower pupil-teacher ratios. Their `rm` values range from approximately 8.034 to 8.780 rooms per dwelling, while their median home values extend from approximately 21.9 to 50.0 thousand dollars.

The unusually high-room suburbs therefore appear to be concentrated among areas with larger homes and generally more favourable housing-market characteristics. However, the relationship is not perfect: at least one high-room suburb has a comparatively high tax rate and a lower house value than the other high-room observations.